

Tax Evasion and Productivity

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Recap

Last meeting:

- ▶ Changes in tax evasion over time in response to fiscal policy changes
- ▶ Feedback: Still working on this
 - ▶ Correct notation
 - ▶ First present results combining all firms, before and after the policy reform
 - ▶ Improve presentation structure to convey a more compelling argument

Today:

- ▶ Tax evasion heterogeneity across firm size



Introduction

- ▶ What's the question?
- ▶ Why should you care?

Contribution

- ▶ What do I do?
- ▶ How do I do it?

Overview of Results

- ▶ What do I find?
- ▶ Why is it important?

Model

Empirical Strategies

- ▶ Testing for the presence of tax evasion by industry
 - ▶ Robust to different specifications
- ▶ Regression
 - ▶ Useful to study tax evasion heterogeneity across firms and over time
- ▶ Deconvolution
 - ▶ Recover distribution of tax evasion and productivity
 - ▶ Estimation of unbiased Production Function

Data

Colombia

Testing

Tax Evasion Over Time

Incentives to evade taxes by cost overreporting changed over time due to changes in fiscal policy

The reform of 1983 increased, from 6% to 10%, the Sales Tax (ST) rate of most industries.

Empirical Approach

Taking the *FOC* and stacking over industries. For Corporations, the truth-reporters, i in industry j at year t , we have,

$$s_{ijt}^C = \gamma_j + \varepsilon_{ijt} \quad (1)$$

Where s_{ijt}^C is the share of materials' cost over sales, γ_j is the output elasticity of materials for industry j (CD-PF), and ε_{ijt} is the error term

Unincorporated Firms

Unincorporated firms, the evaders, we have,

$$s_{ijt}^N = \gamma_j + e_{ijt} + \varepsilon_{ijt} \quad (2)$$

Where e_{ijt} is the overreporting of materials' cost due to tax evasion.

Combining Corporations and Unincorporated Firms

Let D_{ijt}^N be a dummy indicating whether the firm is an unincorporated firm

$$\begin{aligned}(1 - D_{ijt}^N)s_{ijt} + D_{ijt}^Ns_{ijt} &= (1 - D_{ijt}^N)(\gamma_j + \varepsilon_{ijt}) \\ &\quad + D_{ijt}^N\left(\gamma_j + e_{ijt} + \varepsilon_{ijt}\right) \\ s_{ijt} &= \gamma_j + D_{ijt}^Ne_{ijt} + \varepsilon_{ijt}\end{aligned}\tag{3}$$

Taking Expectations

$$E[s_{ijt}|D^N = 0] = \gamma_j \quad (4)$$

$$E[s_{ijt}|D^N = 1] - E[s_{ijt}|D^N = 0] = \mu$$

we can identify the average overreporting level, μ

Changes Over Time

In the case of the fiscal policy changes, we might be interested in the changes in average tax evasion after the 1983 fiscal reform.

We can take the difference of the expectations between year t' and $t = 1983$.

$$\begin{aligned} s_{ijt'} - s_{ijt=1983} &= D_{ijt}^N (e_{ijt'} - e_{ijt=1983}) + (\varepsilon_{ijt'} - \varepsilon_{ijt=1983}) \\ E[s_{ijt'} - s_{ijt=1983} | D^N = 1] &= \mu_{t'} - \mu_{t=1983} \\ &\equiv \Delta_{t'} \end{aligned} \tag{5}$$

μ_t is the annual average tax evasion of year t , and Δ_t is the change in average tax evasion between year t and 1983, the fiscal reform year.

In Regression Form

To estimate Equation 3, I use the following regression specification,

$$s_{ijt} = \gamma_j + \sum_{t=1981}^{1991} \mu_t D_{ijt}^N + \eta_{ijt} \quad (6)$$

- ▶ Where μ_t are year dummies,
 - ▶ μ_t is the average tax evasion of unincorporated firms in year t
- ▶ γ_j , the industry fixed effects, captures the output elasticity of materials of each industry
- ▶ D_{ijt}^N and are defined as before.
- ▶ $\eta_{ijt} = \varepsilon_{ijt} + \xi_{ijt}$, where $e_{ijt} = \sum_t \mu_t + \xi_{ijt}$, and ξ_{ijt} is the idiosyncratic component of tax evasion that is not captured by the year dummies

In Regression Form: Differences Over Time

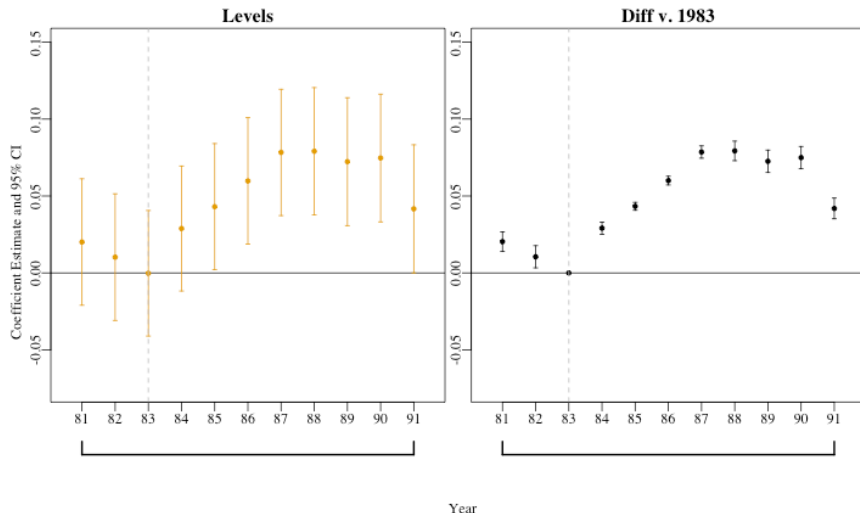
Likewise, the following regression specification is employed to estimate Equation 5,

$$s_{ijt} = \mu D_{ijt}^N + \sum_{t=1981}^{1991} \mathbb{1}[t \neq 1983] \Delta_t D_{ijt}^N + \gamma_j + \eta_{ijt} \quad (7)$$

- ▶ Where Δ_t are year dummies,
 - ▶ Δ_t is the difference in tax evasion of unincorporated firms between year t and 1983, the fiscal reform year
- ▶ μ , the average tax evasion of unincorporated firms in 1983
- ▶ γ_j , D_{ijt}^N and η_{ijt} are defined as before.

Tax Evasion Increases after the Fiscal Reform of 1983

Tax Evasion by Unincorporated Firms Overall Input Overreporting Across All Industries



Results

- ▶ After the fiscal reform of 1983 increased the sales tax rate, tax evasion through input overreporting increased gradually
- ▶ Input overreporting reached an average of 8% in 1987, where it stabilized until 1990

Why Do The Standard Errors Look So Different?

- ▶ **Levels:** The large error terms suggest that tax evasion is widely heterogeneous across firms
 - ▶ Where does this heterogeneity come from? Differences in the tax rate due to firm characteristics (industry, JO, and exports), the (perceived) probability of detection (model assumes risk neutrality, but firms might have different perceived probs), and the cost of evasion across firms.
- ▶ **Differences over time:** Tight standard errors suggest that tax evasion heterogeneity is constant over time.
 - ▶ This might imply that the distributions of the firm characteristics, perceived probability of detection, and cost of evasion are constant over time.

Tax Evasion Over Time (Continued)

The 1983 fiscal reform changed incentives to evade taxes over time *differently* for firms in ST-liable and ST-exempt industries,

- ▶ Most ST-liable industries faced a Sales Tax (ST) rate *increase* from 6% to 10%.
- ▶ ST-exempt industries *remained exempt* before and after the reform.

Note that firms in *all* industries are subject to Corporate Income Tax (CIT).

- ▶ CIT also generates incentives to evade taxes by overreporting input costs.
 - ▶ Lower reported profits due to higher input costs reduce the tax base, and thus the tax liability.
- ▶ Therefore, CIT and ST generate tax evasion incentives for firms in ST-liable industries, while only CIT for firms in ST-exempt industries.

Unincorporated Firms by Industry Tax Liability

For unincorporated firms, the ST and CIT generated different incentives to evade taxes depending on whether they belong to an ST-exempt industry or not,

$$s_{ijt}^N = \gamma_j + e_{ijt}^{ST} + e_{ijt}^{CIT} + \varepsilon_{ijt} \quad j \in \text{ST-Liable} \quad (8)$$

$$s_{ijt}^N = \gamma_j + e_{ijt}^{CIT} + \varepsilon_{ijt} \quad j \in \text{ST-Exempt} \quad (9)$$

where e_{ijt}^{ST} and e_{ijt}^{CIT} stand for the overreporting due to ST and CIT, respectively.

Combining All Firms

- ▶ Let D_{ijt}^E be a dummy indicating whether the industry is ST-exempted; and
- ▶ D_{ijt}^L , a dummy indicating whether the industry is ST-liable
 - ▶ such that $D_{ijt}^E + D_{ijt}^L = 1$.

Unincorporated Firms in ST-Liable and ST-Exempt Industries

Combining Equation 9 with Equation 8,

$$\begin{aligned} D_{ijt}^E s_{ijt}^N + D_{ijt}^L s_{ijt}^N &= D_{ijt}^E (\gamma_j + e_{ijt}^{CIT} + \varepsilon_{ijt}) \\ &\quad + D_{ijt}^L (\gamma_j + e_{ijt}^{ST} + e_{ijt}^{CIT} + \varepsilon_{ijt}) \\ (D_{ijt}^E + D_{ijt}^L) s_{ijt}^N &= (D_{ijt}^E + D_{ijt}^L) \gamma_j + D_{ijt}^E e_{ijt}^{CIT} \\ &\quad + D_{ijt}^L (e_{ijt}^{ST} + e_{ijt}^{CIT}) + (D_{ijt}^E + D_{ijt}^L) \varepsilon_{ijt} \\ s_{ijt}^N &= \gamma_j + D_{ijt}^E e_{ijt}^{CIT} + D_{ijt}^L (e_{ijt}^{ST} + e_{ijt}^{CIT}) + \varepsilon_{ijt} \end{aligned} \tag{10}$$

Corporations and Unincorporated Firms

Likewise, combining Equation 1 with Equation 10,

$$\begin{aligned}(1 - D_{ijt}^N)s_{ijt} + D_{ijt}^N s_{ijt} &= (1 - D_{ijt}^N)(\gamma_j + \varepsilon_{ijt}) \\ &\quad + D_{ijt}^N \left(\gamma_j + D_{ijt}^E e_{ijt}^{CIT} \right. \\ &\quad \left. + D_{ijt}^L (e_{ijt}^{ST} + e_{ijt}^{CIT}) + \varepsilon_{ijt} \right) \quad (11) \\ s_{ijt} &= \gamma_j + D_{ijt}^N D_{ijt}^E e_{ijt}^{CIT} \\ &\quad + D_{ijt}^N D_{ijt}^L (e_{ijt}^{ST} + e_{ijt}^{CIT}) + \varepsilon_{ijt}\end{aligned}$$

Taking Expectations

$$E[s_{ijt}|D^N = 0] = \gamma_j$$

$$E[s_{ijt}|D^N = 1, D^E = 1, D^L = 0] - E[s_{ijt}|D^N = 0] = \mu_E^{CIT}$$

$$E[s_{ijt}|D^N = 1, D^E = 0, D^L = 1] - E[s_{ijt}|D^N = 0] = \underbrace{\mu_L^{ST} + \mu_L^{CIT}}_{\mu_L^{ST+CIT}}$$

we can identify the average overreporting by firms in ST-exempt, μ_E^{CIT} , and in ST-liable industries, μ_L^{ST+CIT}

If we assume $\mu_E^{CIT} = \mu_L^{CIT}$, we can also identify the overreporting due to ST, μ_L^{ST} .

Changes Over Time

In the case of the fiscal policy changes, we might be interested in the changes in average tax evasion after the 1983 fiscal reform.

We can take the difference of the expectations of year t' after the reform and the reform year, $t = 1983$,

$$\begin{aligned} E[s_{ijt'} - s_{ijt=1983} | D^N = 1, D^E = 1, D^L = 0] &= \mu_{E,t'}^{CIT} - \mu_{E,t=1983}^{CIT} \\ &\equiv \Delta_{E,t'}^{CIT} \end{aligned}$$

$$\begin{aligned} E[s_{ijt'} - s_{ijt=1983} | D^N = 1, D^E = 0, D^L = 1] &= \mu_{L,t'}^{CIT+ST} - \mu_{L,t=1983}^{CIT+ST} \\ &\equiv \Delta_{L,t'}^{CIT+ST} \end{aligned}$$

In Regression Form

To estimate Equation 11, I use the following regression specification,

$$s_{ijt} = \gamma_j + \sum_{t=1981}^{1991} \mu_{E,t} D_{ijt}^N D_{ijt}^E + \sum_{t=1981}^{1991} \mu_{L,t} D_{ijt}^N D_{ijt}^L + \eta_{ijt} \quad (12)$$

- ▶ $\mu_{E,t}$ are the coefficients of input overreporting by firms in ST-exempt in year t , and $\mu_{L,t}$ of firms in ST-liaible industries
- ▶ γ_j , D_{ijt}^N , D_{ijt}^E , D_{ijt}^L , and η_{ijt} are defined as before.

Differences Over Time in Regression Form

$$\begin{aligned}s_{ijt} = & \gamma_j + \mu_0 D_{ijt}^N + \mu_1 D_{ijt}^N D_{ijt}^E + \mu_2 D_{ijt}^N D_{ijt}^L \\ & + \sum_{t=1981}^{1991} \mathbb{1}[t \neq 1983] \Delta_{E,t} D_{ijt}^N D_{ijt}^E \\ & + \sum_{t=1981}^{1991} \mathbb{1}[t \neq 1983] \Delta_{L,t} D_{ijt}^N D_{ijt}^L + \eta_{ijt}\end{aligned}\tag{13}$$

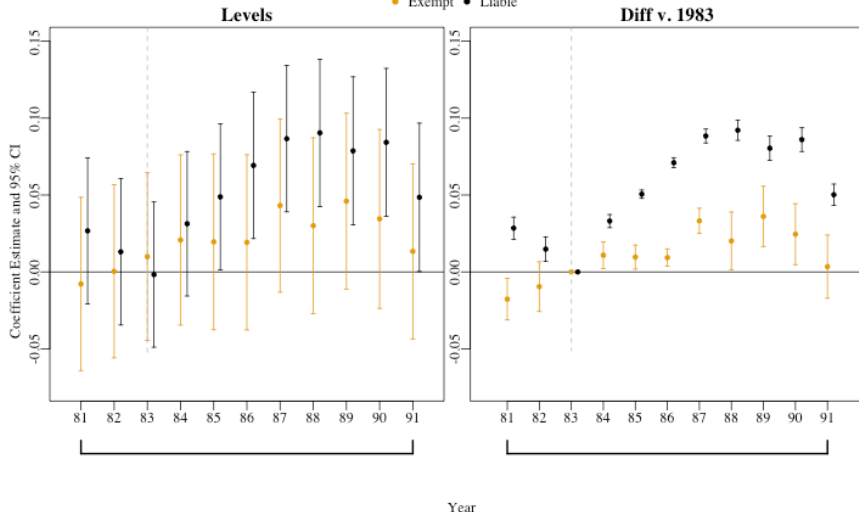
- ▶ $\Delta_{E,t}$ are the coefficients of the change in average tax evasion of firms in ST-exempt relative to 1983 and $\Delta_{L,t}$ of firms in ST-liable industries
- ▶ μ_0 , μ_1 , and μ_2 capture the average tax evasion of firms in 1983 of firms in both ST-liable and ST-exempt industries
- ▶ γ_j , D_{ijt}^N , D_{ijt}^E , D_{ijt}^L , and η_{ijt} are defined as before.

Firms Facing Larger Incentives Evade More

Tax Evasion by Unincorporated Firms

Input Overreporting in ST-Exempt and ST-Liable Industries

● Exempt ● Liable



Unincorporated Firms in Both Types of Industries

After the fiscal reform of 1983, we observe:

- ▶ Firms in ST-liable industries increased tax evasion gradually until 1987, where it stabilized until 1990
- ▶ Firms in ST-exempt industries show a moderate but significant increase in tax evasion after the fiscal reform of 1983, it then increased again between 1987 and 1990
- ▶ The increase is *larger* for ST-liable firms, the most affected by the fiscal reform of 1983

Splitting Unincorporated Firms by JO into LLCs and Proprietorships

Changes in Tax Evasion Over Time by Industry Type and Juridical Organization (JO)

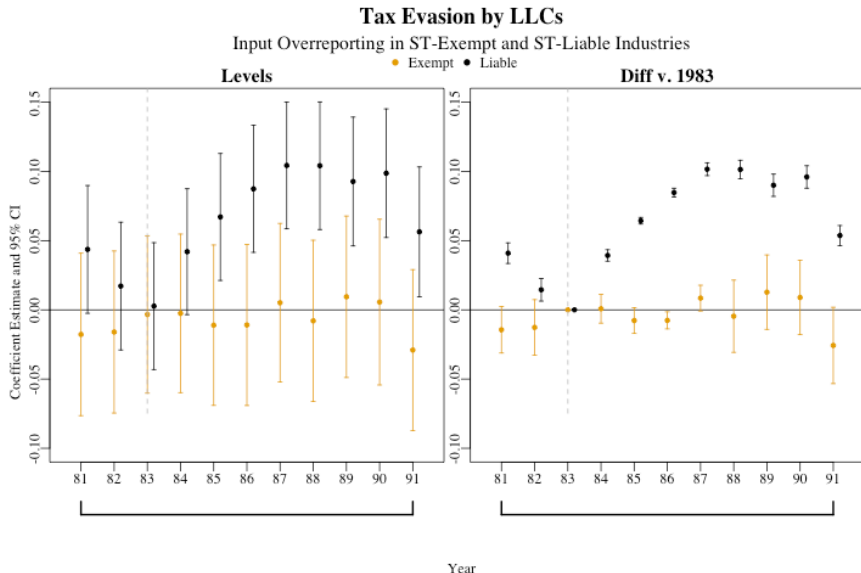
The tax evasion incentives generated by the 1983 fiscal policy reform also differed by JO,

- ▶ Differences *across* industries:
 - ▶ Most ST-liable industries faced a Sales Tax (ST) rate *increase* from 6% to 10%.
 - ▶ ST-exempt industries before the reform *remained exempt* after the reform.
- ▶ Differences *across* juridical organizations (JO) *within* industries:
 - ▶ Corporate Income Tax (CIT) increased 8% on average for most Proprietorships (PRTs) (through Individual Income Tax), while
 - ▶ CIT decreased 2 percentage points, from 20% to 18%, for Limited Liability Companies (LLCs)

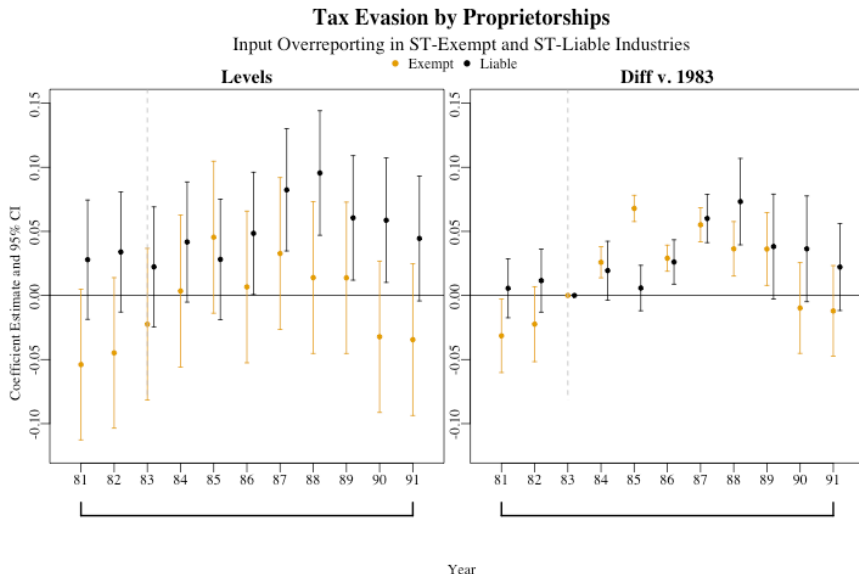
Expected Changes in Tax Evasion After the Fiscal Reform of 1983

- ▶ Since the **ST** rate *increased* for most ST-liable industries, we expect an *increase* in average *tax evasion* by these firms
- ▶ With respect to **CIT**, the expected change is *ambiguous*
 - ▶ Proprietorships faced an *increase* in their incentives to evade taxes, while
 - ▶ LLCs a small *decrease*.

LLCs with Larger Incentives Evade More



Proprietorships Response is More Heterogeneous



LLCs vs Proprietorships

- ▶ LLCs in ST-liable industries show a significant increase in tax evasion after the fiscal reform of 1983, consistent with the increase in their incentives to evade taxes due to the increase in the ST rate
- ▶ LLCs in ST-exempt industries show no significant change, suggesting that the small decrease in CIT for LLCs did not generate a significant decrease in tax evasion

LLCs vs Proprietorships (Cont.)

- ▶ Proprietorships in ST-exempt industries show a moderate but significant increase in tax evasion, consistent with the increase in their incentives to evade taxes due to the increase in CIT
- ▶ Proprietorships in ST-liable industries show no immediate change in tax evasion (1984-1985), but an increase between 1986 and 1989
- ▶ Overall, Proprietorships show a more heterogeneous response, as shown by the larger standard errors.
 - ▶ This makes sense, Proprietorships are likely to be more heterogeneous in characteristics (grandpa's workshop vs. young entrepreneurs, varying levels of process standardization, etc)

Takeaways

- ▶ Evidence suggests firms increase tax evasion by overreporting their inputs (materials) in response to increases in CIT and ST
- ▶ Overall, firms facing *larger increases* in ST and CIT tax rates evade more by overreporting a larger share of their input costs
- ▶ Firms facing a decrease in tax rates, such as LLCs in ST-exempt industries, show no significant decrease in tax evasion.
 - ▶ This finding suggests that firms might respond *asymmetrically* to changes in their incentives to evade taxes.

Tax Evasion and Firm Heterogeneity

Tax Evasion and Size

- ▶ Are there systematic differences in tax evasion across firms with different characteristics?
- ▶ Why? Firms might face different probabilities of detection, cost of evading taxes, and even different tax rates —thus, different incentives— depending on their characteristics.
- ▶ Size in particular:
 - ▶ Common intuition is that larger firms face higher probabilities of detection, and thus evade less.
 - ▶ However, recent evidence suggests that tax evasion increases along the size distribution (revenue) and then drops at the top of the distribution (Carrillo et al. 2022).

Empirical Strategy

Starting from FOC, as before, for corporations, the truth-reporters, we have,

$$s_{ijt}^C = \gamma_j + \varepsilon_{ijt}$$

For unincorporated firms, the evaders, we now have,

$$s_{ijt}^N = \gamma_j + e_{ijt}(\tau, q, \kappa) + \varepsilon_{ijt} \quad (14)$$

- ▶ where the tax evasion, $e_{ijt}(\tau, q, \kappa)$, is a function of
 - ▶ the effective sales tax rate τ ,
 - ▶ the probability of detection q and
 - ▶ the cost of evasion κ .
- ▶ γ_j and ε_{ijt} are defined as before.

Empirical Strategy (Cont.)

Combining the two types of firms, we have,

$$\begin{aligned}(1 - D_{ijt}^N)s_{ijt} + D_{ijt}^Ns_{ijt} &= (1 - D_{ijt}^N)(\gamma_j + \varepsilon_{ijt}) \\ &\quad + D_{ijt}^N(\gamma_j + e_{ijt}(\tau, q, \kappa) + \varepsilon_{ijt}) \quad (15) \\ s_{ijt} &= \gamma_j + D_{ijt}^Ne_{ijt}(\tau, q, \kappa) + \varepsilon_{ijt}\end{aligned}$$

► where D_{ijt}^N is defined as before

Empirical Strategy (Cont.)

- ▶ Since tax evasion $e_{ijt}(\cdot)$, the probability of detection q , and the cost of evasion κ are all **unobserved**, we could use a flexible function of **firm characteristics**, x_{ijt} , that are likely to influence them
- ▶ The error terms is theoretically independent of most firm characteristic that could influence tax evasion, $E[\varepsilon_{ijt}|x_{ijt}] = 0$ and $E[e_{ijt}|x_{ijt}] \neq 0$.
- ▶ Current revenue however is not independent of the error term, as it is part of the firm's output, and thus is correlated with the error term of the production function.
- ▶ Nevertheless, I can learn the expected value of the error term conditional on the firm's characteristics, $E[\varepsilon_{ijt}|x_{ijt}]$, from the Corporations.

Taking Expectations

Taking expectation of Equation 15,

$$\begin{aligned} E[s_{ijt}|D^N = 0, x_{ijt}] &= \gamma_j + E[\varepsilon_{ijt}|D^N = 0, x_{ijt}] \\ &= \gamma_j + E[\varepsilon_{ijt}|x_{ijt}] \\ E[s_{ijt}|D^N = 1, x_{ijt}] &= \gamma_j + E[e_{ijt}|D^N = 1, x_{ijt}] + E[\varepsilon_{ijt}|D^N = 1, x_{ijt}] \\ &= \gamma_j + E[e_{ijt}|D^N = 1, x_{ijt}] + E[\varepsilon_{ijt}|x_{ijt}] \end{aligned}$$

where the second and fourth line follow from the independence of the error term across Corporations and Unicorporated firms, $E[\varepsilon_{ijt}|D^N, x_{ijt}] = E[\varepsilon_{ijt}|x_{ijt}]$.

In addition, when x_{ijt} is not current revenue, then $E[\varepsilon_{ijt}|x_{ijt}] = 0$ holds.

Taking Expectations (Cont.)

Finally,

$$\begin{aligned} E[s_{ijt}|D^N = 1, x_{ijt}] - E[s_{ijt}|D^N = 0, x_{ijt}] &= E[e_{ijt}(\tau, q, \kappa)|D^N = 1, x_{ijt}] \\ &\equiv \mu_x(\tau, q, \kappa) \end{aligned}$$

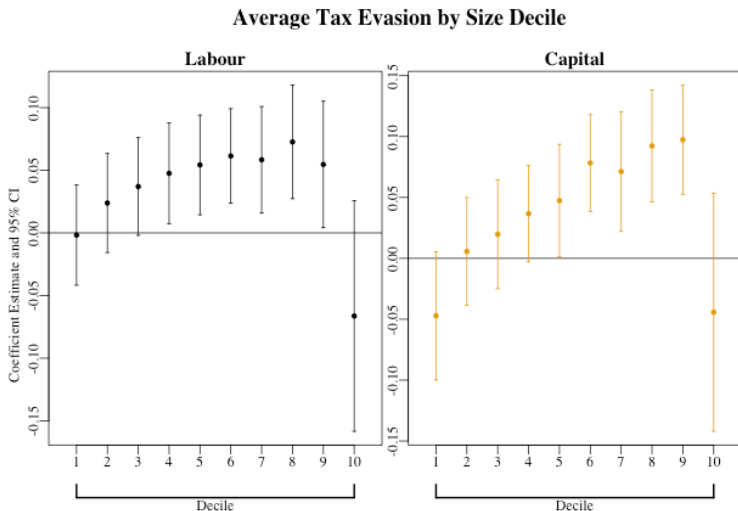
Regression Specification

To estimate Equation 15, I use ,

$$s_{ijt} = \gamma_j + D_{ijt}^N \sum_{h=1}^{10} \mathbb{1}[x_{ijt} \in \mathcal{X}_h] \mu_h + \sum_{h=1}^{10} \mathbb{1}[x_{ijt} \in \mathcal{X}_h] \lambda_h + \eta_{ijt} \quad (16)$$

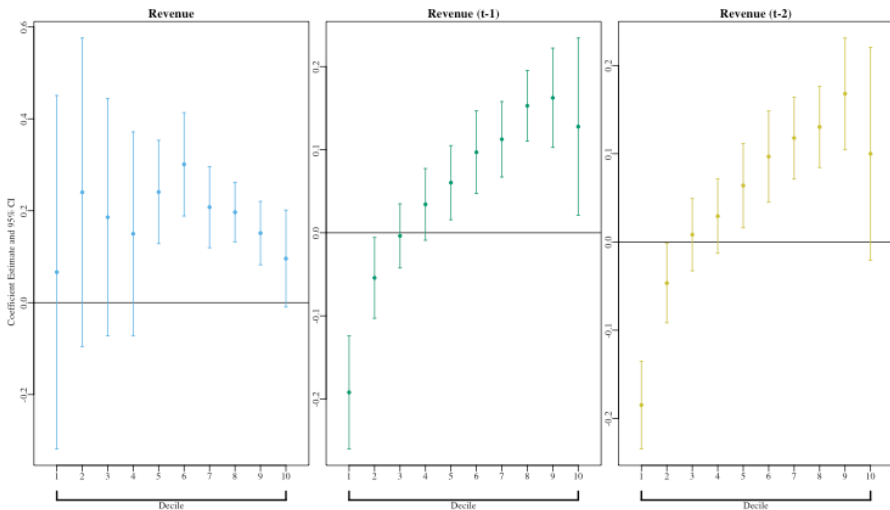
- ▶ x_{it} are the firm's decile of size measured as either labor, capital, or revenue (current, $t - 1$, and $t - 2$)
 - ▶ μ_h , the coefficients and capture the average tax evasion of firms in each decile of size
- ▶ λ_h captures $E[\varepsilon_{ijt}|x_{ijt}]$, only included when x_{ijt} is current revenue
- ▶ γ_j , industry fixed effects, and η_{ijt} the error term, includes the component of tax evasion that is not captured by the size deciles, and ε_{ijt} when x_{ijt} is not current revenue

Tax Evasion Increases Along the Size Distribution and Then Drops at the Top



Tax Evasion Increases Along the Size Distribution and Then Drops at the Top

Average Tax Evasion by Size Decile



decile	Capital	Labour	Revenue	Revenue _{t-1}	Revenue _{t-2}
1	-0.0168	0.0077	0.0312	-0.1485***	-0.1334***
	[-0.0424, 0.0061]	[-0.0141, 0.0291]	[-0.2914, 0.5099]	[-0.1744, -0.1232]	[-0.1677, -0.1081]
2	0.0411***	0.0361***	0.2382*	-0.0138	0.0027
	[0.0147, 0.0648]	[0.0162, 0.0574]	[-0.0247, 0.6498]	[-0.0385, 0.0094]	[-0.0211, 0.0342]
3	0.0412***	0.0424***	0.1724**	0.0321***	0.0473***
	[0.0103, 0.0666]	[0.0197, 0.0624]	[0.0121, 0.4101]	[0.009, 0.0561]	[0.0218, 0.0714]
4	0.055***	0.0582***	0.1273	0.0694***	0.0709***
	[0.0306, 0.0816]	[0.0373, 0.0788]	[-0.0388, 0.288]	[0.0448, 0.0924]	[0.0459, 0.0972]
5	0.0598***	0.0686***	0.2445***	0.0876***	0.0961***
	[0.0382, 0.0909]	[0.0464, 0.0914]	[0.1851, 0.3845]	[0.0649, 0.1111]	[0.0702, 0.1199]
6	0.0849***	0.0693***	0.3083***	0.1169***	0.1216***
	[0.0607, 0.1087]	[0.0498, 0.0872]	[0.2604, 0.4254]	[0.0982, 0.1434]	[0.1006, 0.1555]
7	0.0703***	0.0664***	0.2131***	0.1265***	0.1401***
	[0.0431, 0.094]	[0.0354, 0.0921]	[0.174, 0.3108]	[0.1044, 0.156]	[0.1133, 0.1703]
8	0.0828***	0.085***	0.1884***	0.1605***	0.1419***
	[0.0506, 0.104]	[0.0534, 0.1087]	[0.1563, 0.2641]	[0.1482, 0.2003]	[0.1301, 0.1935]
9	0.0816***	0.0622***	0.1339***	0.151***	0.1656***
	[0.0579, 0.1239]	[0.0376, 0.1157]	[0.1281, 0.2107]	[0.1332, 0.1986]	[0.1448, 0.2128]
10	-0.0186	-0.0328	0.106***	0.145***	0.1267***
	[-0.0986, 0.049]	[-0.0974, 0.0158]	[0.0541, 0.1826]	[0.0771, 0.1988]	[0.0516, 0.1888]

Tax Evasion and Size

- ▶ Similar pattern across size measures: Tax evasion increases along the size distribution and then drops at the top of the distribution
- ▶ Surprisingly consistent with recent evidence in the literature using more detailed tax admin data (Figure 1).
- ▶ Large firms evade more than small firms, but the largest firms do not evade taxes.

But Why?

- ▶ Since the probability of detection cannot be, by logic, decreasing in size,
 - ▶ Most likely it is flat along the size distribution, and then increases rapidly at the top of the distribution
- ▶ It then follows that the cost of evasion should be decreasing along the size distribution, and then it might (or might not) increase at the top of the distribution.

Why the Negative Sign for Some Small Firms?

- ▶ Smaller firms are more likely to be Proprietorships and less formal
- ▶ Proprietorships report their industrial activity as part of their individual income taxes. They need to keep track of their personal expenses and their industrial activity expenses
- ▶ Because they are more likely to be less formal than larger firms, they may lack a rigorous expense report system
- ▶ Thus, they might be consistently underreporting their expenses

Marginal Effects

All previous measures were correlated with size,

- ▶ Is the pattern observed in the previous section explained by one of the size measures in particular?
- ▶ What is the marginal effect of each size measure holding the others constant?

Regression Specification

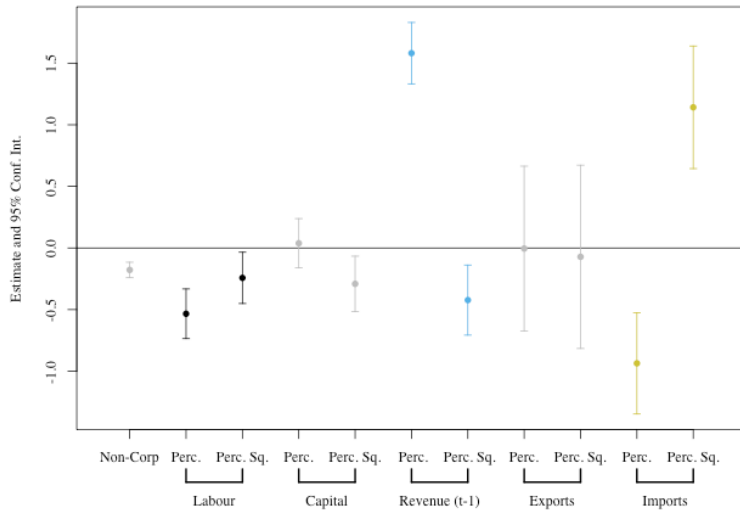
I use the following regression specification of Equation 15 to investigate these questions,

$$s_{ijt} = \gamma_j + \mu_0 D_{ijt}^N + D_{ijt}^N \sum_{z \in \mathcal{Z}} \sum_{h \in \{1,2\}} \mu_h^z (x_{ijt}^z)^h + \eta_{ijt}$$

- ▶ where now x_{it} stands for the firm's **percentile** of $\mathcal{Z} = \{\text{labour, capital, lagged revenue, exports, imports}\}$, including the square term, $h = 2$, to capture non-linearities,
- ▶ μ_h^z , the corresponding coefficients
 - ▶ the marginal effect of size measure z is given by $\frac{\partial s_{ijt}}{\partial x_{ijt}^z} = \mu_1^z + 2\mu_2^z (x_{ijt}^z)$
- ▶ γ_j and η_{ijt} same as before

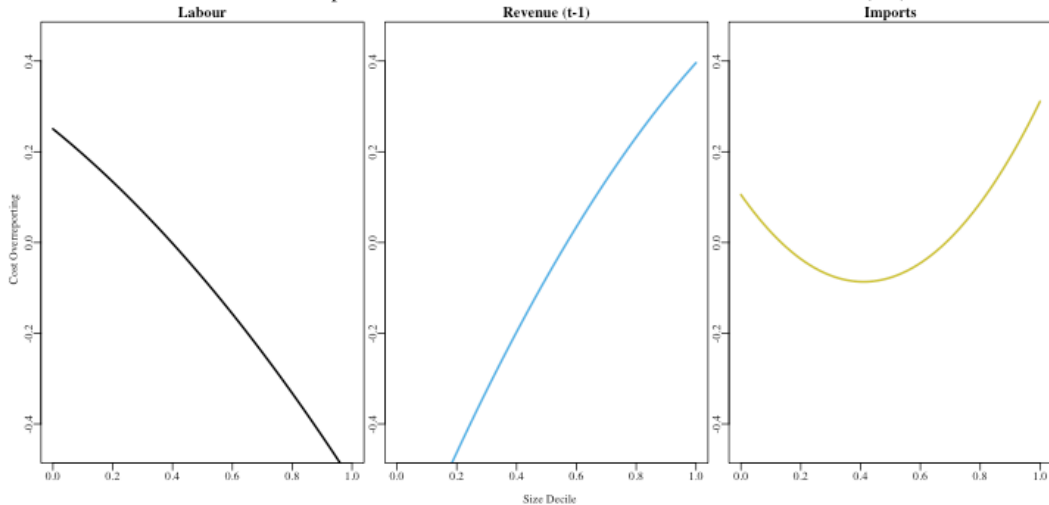
Tax Evasion and Firm Size

Conditional Effects of Size on Tax Evasion



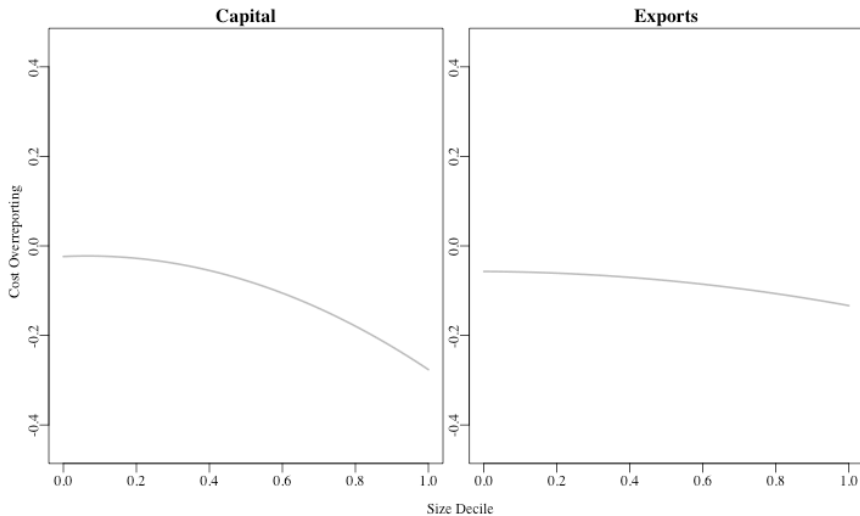
Expected Tax Evasion

Conditional Expectation Function. Other Variables Fixed at The Median (0.5)



Expected Tax Evasion

Conditional Expectation Function. Other Variables Fixed at The Median (0.5)



Marginal Effects of Different Size Measures

Holding other size measures constant, as firms grow in

- ▶ **labour**, tax evasion **decreases** at a marginal increasing rate (decreasing and concave) as firms grow
- ▶ **revenue** _{$t-1$} , tax evasion **increases** at a marginal decreasing rate (increasing and concave) as firms grow
- ▶ **imports**, tax evasion **decreases** at a marginal decreasing rate (decreasing and convex) as firms grow

Does This Make Sense? Labour

Comparing two firms with similar characteristics but different percentiles of labour, the firm with a *higher* percentile of labour *evades less*

- ▶ As firms grow in labour, they face *higher probabilities of detection*, as the probability of being denounced to the tax authority by an employee increases
- ▶ They might also face *higher costs of evading taxes* as more employees need to be involved in the evasion scheme (e.g. larger accounting department and more complex operations, keeping double accounting books, warehouse employees might also need to be involved, etc.).
 - ▶ So the time and resources these employees need to allocate to tax evasion, instead of productive work, increases, and thus the cost of evading taxes increases as firms grow in labour.

Revenue

Comparing two firms with similar characteristics but different percentiles of revenue (t-2), the firm with a *higher* percentile of revenue *evades more*

- ▶ As firms grow in revenue, they have more resources to allocate to tax evasion, and thus face *lower costs of evading taxes relatively to low revenue firms*
- ▶ The *cost of evasion* might *abruptly increase for the largest firms*.
 - ▶ The largest firms might be able to implement more sophisticated but *legal tax avoidance* strategies (e.g. profit offshoring), and thus the cost of evading taxes *illegally* increases significantly relative to legal schemes
 - ▶ In 2016 during his presidential campaign, Donald Trump famously admitted avoiding paying income tax for years and that he was *smart* for doing so (**Barstow2016?**).
 - ▶ The reputational cost of being caught evading taxes might be higher for the largest firms, as they are more visible to the public and potential investors.

Revenue (Cont.)

- ▶ Larger revenue might also increase the *probability of detection*, as visibility increases
 - ▶ The evidence suggests that the increase in the detection probability due to larger revenue should be relatively smaller than the decrease in the evasion cost for most of the size distribution
- ▶ However, at the *top of the revenue distribution*, the probability of detection might *increase significantly* as it becomes strategically targeted by the tax authority, and thus tax evasion drops for the largest firms

Imports

- ▶ Imports in Colombia are subject to ST on top of the regular import duties.
- ▶ Imports are also harder to misreport because they leave larger paper trails, and thus the authority can more easily trace them.
- ▶ Consequently, a larger share of imports increases the incentives to evade taxex by overreporting local input costs
 - ▶ larger share of imports increases the share of total material costs that cannot be claimed as deductions from ST, resulting in lower net after-tax profits, and thus increasing incentives for tax evasion

Imports (Cont.)

This explains why firm in the highest percentiles of imports evade more

- ▶ At lower levels (below the 40th perc.), incentives are small. Evasion looks like it decreases with imports at low levels
- ▶ As firms move above the 40th percentile however, the incentives to evade taxes increase significantly

Takeaways

Overall, these results suggest that the non-linear pattern observed in the previous section might be explained by:

1. a probability of detection that remains flat (or increases slowly) along most of the size distribution, but then increases sharply for the largest firms.
 - ▶ The detection probability increases sharply for the largest firms because they are more visible to the tax authority, and because
 - ▶ these firms are more likely to be denounced to the tax authority by an employee

Takeaways (Cont.)

2. a cost of evasion that decreases along the size distribution (but then it might increase rapidly for the largest firms). This is because
 - ▶ firms with small revenues face higher relative costs of evasion as they have fewer resources to allocate to tax evasion. As firms increase in revenue, they can allocate more resources to tax evasion, reducing the relative cost of evasion.
 - ▶ For the largest revenue firms however, the cost of evasion might increase abruptly due to the higher costs of illegal evasion relative to legal avoidance strategies accessible mostly for the largest firms only, and higher reputational costs if caught evading taxes.

Appendix

Appendix

B) Value of Reported Ghost Purchases Over Firm Total Purchases, by Firm Size

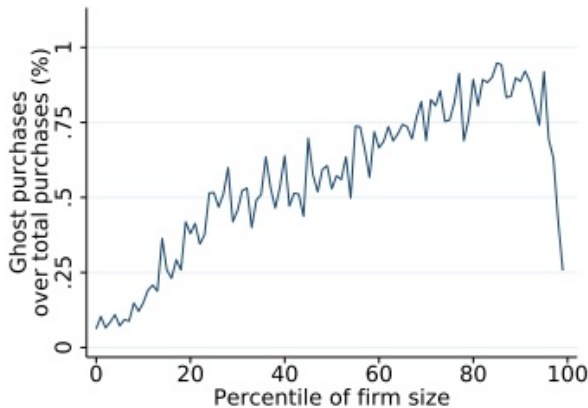
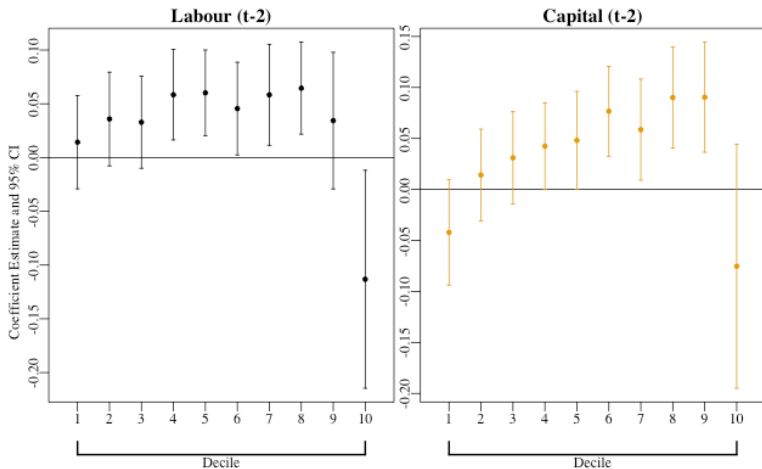


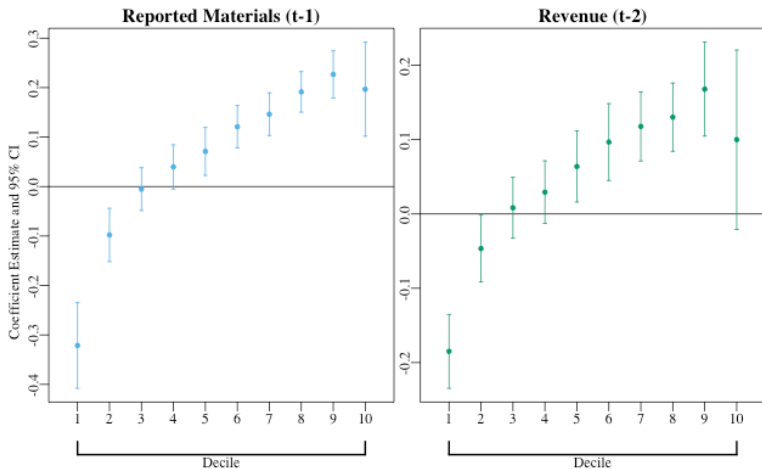
Figure 1: Figure 1B from Carrillo et al. (2022) ↵

Robust Check for Size Heterogeneity

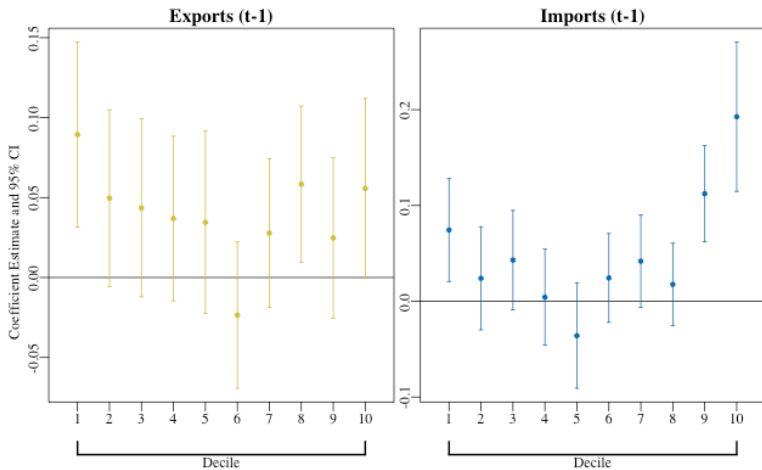
Average Tax Evasion by Size Decile



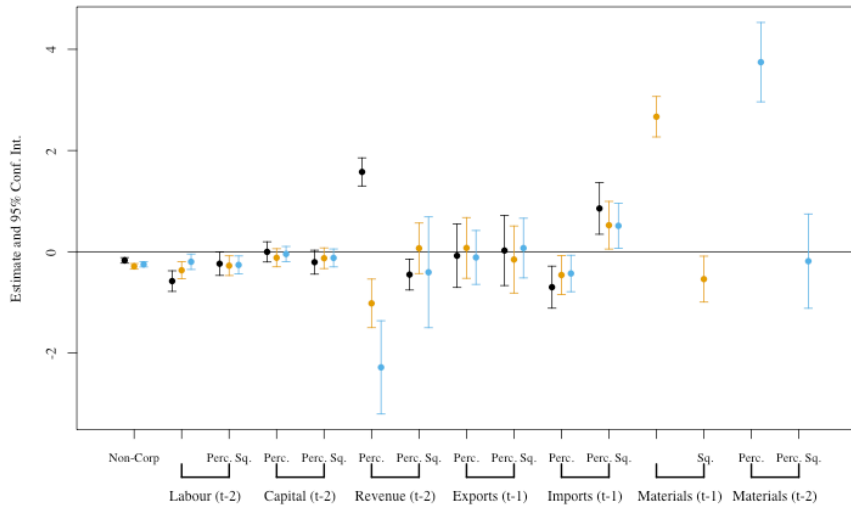
Average Tax Evasion by Size Decile



Average Tax Evasion by Size Decile



Tax Evasion and Size: Marginal Effects



References

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